

iQOO 15 Edge-AI Compute: Deep Technical Assessment for Your Phone-First Engineering Agent

Executive finding

After re-checking the iQOO 15 against **iQOO's own current specifications, Qualcomm's Snapdragon 8 Elite Gen 5 documentation, Qualcomm AI Hub's current 2026 runtime/model support, and the iQOO Hackathon material**, the central conclusion from the earlier analysis holds — but it can now be made much more precise.

The iQOO 15 is genuinely capable of being the **local intelligence layer** of your system rather than merely a remote-control screen for a cloud agent. The phone combines an 8-core Snapdragon 8 Elite Gen 5, Hexagon NPU, Adreno GPU, 12/16 GB RAM configurations, UFS 4.1, a 7000 mAh battery, substantial vapor-chamber cooling, Wi-Fi 7, and Android 16/OriginOS 6. iQOO's actual CPU configuration is the 4.6 GHz Snapdragon variant: **2 × 4.6 GHz + 6 × 3.62 GHz.** ¹

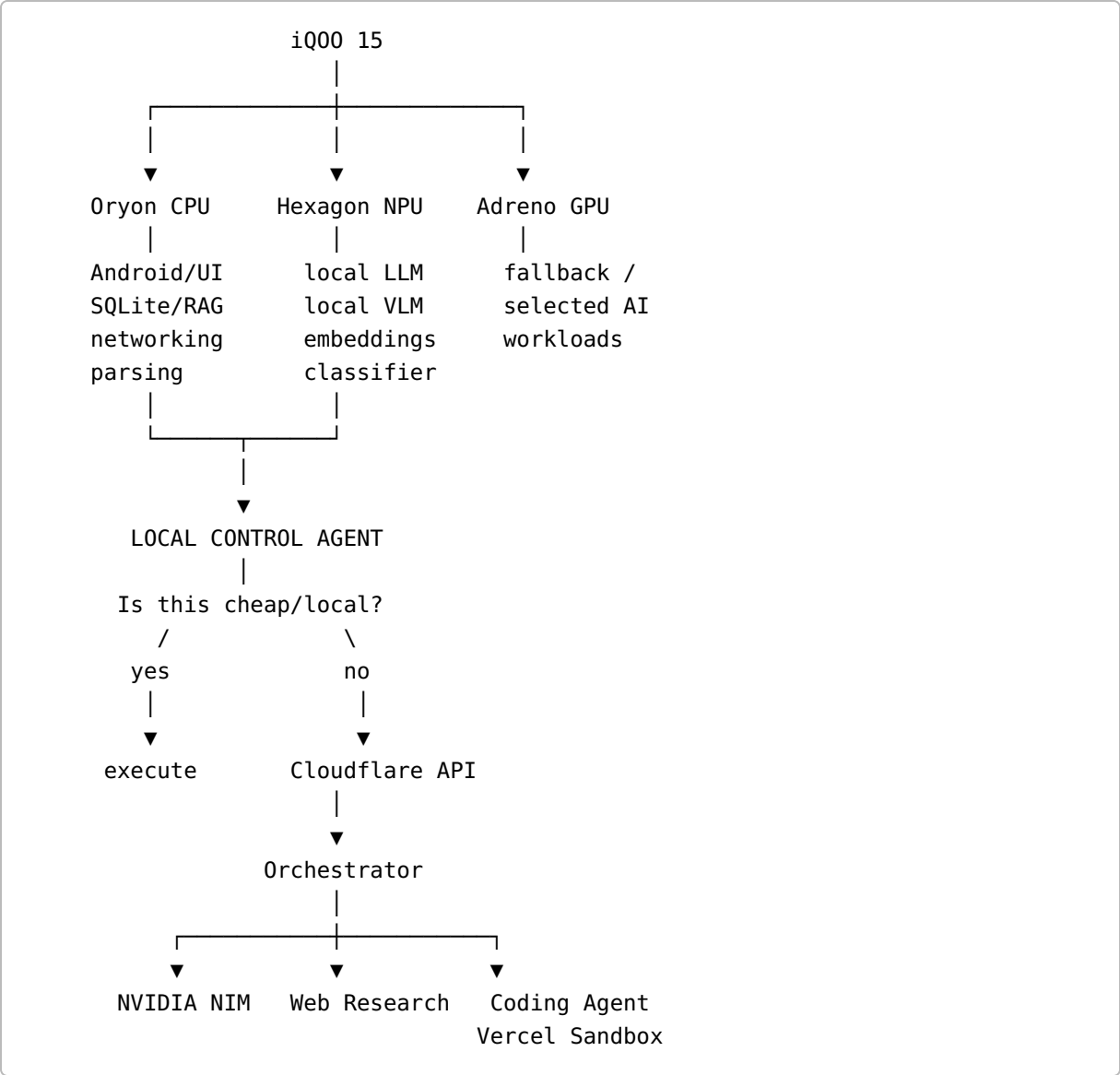
More importantly, the software story is now substantially better than simply saying “Snapdragon has an NPU.” Qualcomm currently provides **GenieX**, an on-device generative-AI runtime capable of targeting Hexagon NPU, Adreno GPU and CPU, with a QAIPT path optimized for the NPU and a llama.cpp/GGUF path for broader model compatibility. Qualcomm AI Hub currently lists **Qwen3-4B, Qwen3-8B, and Qwen3-VL-4B-Instruct as supporting Snapdragon 8 Elite Gen 5 Mobile**, which provides much stronger evidence that 4B–8B local models are technically viable on this chipset than a generic parameter-count estimate alone. ²

That does **not** mean I would architect your hackathon project around an 8B model. For your particular product, my recommendation after the deeper research is:

Make a hardware-optimized ~4B local model the primary on-device reasoning engine, use smaller specialist models where appropriate, and treat 8B as an optional demonstration/quality tier rather than the minimum requirement.

That gives you enough local intelligence for intent understanding, task routing, structured actions, diff explanation, local project memory, risk classification, notification summarization, and potentially screenshot reasoning, while retaining enough RAM, thermal and latency margin for your Android UI, networking, database/cache and background services. Qualcomm's current AI Hub explicitly identifies Qwen3-4B and Qwen3-VL-4B-Instruct as hardware-optimized models and lists the Snapdragon 8 Elite Gen 5 chipset among their supported targets. ³

The ideal architecture therefore becomes:



That is a defensible **hybrid edge/cloud AI architecture**, rather than an artificial claim that a phone should perform entire-repository coding or run a massive frontier model locally. Qualcomm itself positions this generation around on-device agentic AI, and the iQOO Hackathon series is explicitly framed as phone-first development. ⁴

Verified iQOO 15 compute platform

There are several places where the earlier specification list can now be tightened.

Component	Verified iQOO 15 capability	Relevance to your system
SoC	Snapdragon 8 Elite Gen 5	Main compute platform
Process	3 nm	Efficiency / thermal headroom

Component	Verified iQOO 15 capability	Relevance to your system
CPU	8-core Qualcomm Oryon	Android logic, retrieval, preprocessing, fallback inference
iQOO CPU clocks	2 × 4.6 GHz + 6 × 3.62 GHz	Strong CPU-side execution
AI accelerator	Qualcomm Hexagon NPU	Primary local AI accelerator
GPU	Qualcomm Adreno	Graphics and supported compute/inference paths
Extra chip	iQOO Supercomputing Chip Q3	Primarily gaming graphics acceleration
RAM	12 GB / 16 GB	Main local-model capacity constraint
Storage	256 GB / 512 GB India configs	Model files, local indexes, caches
Storage type	UFS 4.1	Fast application/model loading
OS	OriginOS 6 based on Android 16	Current Android development base
Battery	7000 mAh typical, 26.25 Wh	Long demo/edge-compute operating window
Cooling	8K single-layer VC system	Sustained workload thermal headroom
Wi-Fi	Wi-Fi 7/6/5	Fast cloud escalation
Bluetooth	Bluetooth 6.0	Peripheral possibilities

iQOO's India specification page confirms the Snapdragon 8 Elite Gen 5, 12 GB and 16 GB RAM options, 256/512 GB storage, 7000 mAh battery, and OriginOS 6 based on Android 16. The global specification page supplies the exact 8-core layout and clock configuration of **2 × 4.6 GHz + 6 × 3.62 GHz**, as well as UFS 4.1.

1

The CPU figure needed one important clarification

Qualcomm's reference Snapdragon specification lists a maximum **4.74 GHz** Oryon CPU configuration, but Qualcomm explicitly notes that a **4.6 GHz variant exists and OEMs choose the platform configuration**. iQOO's own device specification identifies its iQOO 15 as the 4.6 GHz implementation. Therefore, for your presentation, use **4.6 GHz for the actual iQOO 15**, not 4.74 GHz. ⁵

Qualcomm claims the third-generation Oryon CPU improves CPU performance by 20% and CPU power efficiency by 35% compared with the preceding Snapdragon generation, while the platform also introduces hardware matrix acceleration on the CPU as part of its AI Engine. Those are vendor generation-over-generation claims rather than independent iQOO 15 benchmarks, so they should be presented as Qualcomm claims rather than your measured results. ⁶

This matters because the CPU can take responsibility for substantial parts of your app even while the NPU performs inference:

CPU

- |
- ├─ Android application
- ├─ Jetpack Compose UI
- ├─ network requests
- ├─ Supabase/WebSocket handling
- ├─ local SQLite/database
- ├─ project/task retrieval
- ├─ tokenization/preprocessing
- ├─ JSON validation
- ├─ cryptographic signing
- ├─ policy logic
- └─ inference fallback

So your architecture should not treat the NPU as the only useful compute resource.

RAM is strong, but the 9.6 Gbps claim needs careful wording

iQOO markets LPDDR5X Ultra at “up to 9.6 Gbps” and UFS 4.1 storage. The India page specifically notes that the tested 16 GB + 512 GB model uses LPDDR5X Ultra at up to 9600 Mbps, while the global page identifies LPDDR5X Ultra on the regular configurations and LPDDR5X Ultra Pro on its 1 TB configuration. ⁷

I would **not** describe 9.6 Gbps as the phone's total memory bandwidth. It is a memory data-rate specification used in iQOO's marketing, not the same thing as aggregate GB/s bandwidth across the complete memory interface. For your pitch, simply say:

12–16 GB high-speed LPDDR5X-class memory

That statement is both accurate and relevant.

The Q3 chip is useful — but don't sell it as your general-purpose AI accelerator

iQOO describes its Supercomputing Chip Q3 as a **mobile graphics chip** containing rendering, ray-tracing and “AI trio core” capabilities, with its published use cases centered on gaming: super-resolution, frame interpolation, ray tracing and display enhancement. ⁸

There is currently no public evidence in the material reviewed that an ordinary third-party Android application can use Q3 as a general-purpose neural-network accelerator in the same way that Qualcomm exposes the Hexagon NPU through QAIRT.

Therefore:

DON'T SAY

"Our AI runs on iQOO's Q3 chip."

SAY

"Our application runs local AI on
Snapdragon's Hexagon NPU.

The iQOO 15 additionally includes Q3,
a dedicated graphics accelerator."

That distinction will save you from a technically informed judge asking for the Q3 developer API.

Cooling and battery are genuinely substantial

iQOO describes the iQOO 15 as using its largest vapor chamber to date, with an 8K VC cooling system, graphene material, dual-layer graphite and thermal gel. Its specification page gives the battery as **7000 mAh typical**, while the detailed footnote specifies **26.25 Wh typical energy**. ⁸

The cooling marketing is primarily validated by iQOO against gaming workloads, not specifically sustained local LLM inference. Therefore the defensible statement is:

The large battery and cooling system give the device **promising thermal and energy headroom for sustained edge-AI workloads, but sustained LLM performance needs to be measured on the actual hackathon handset.**

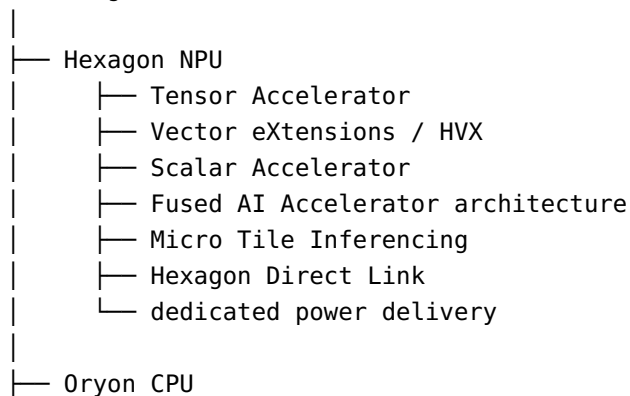
That is stronger technically than claiming the VC automatically guarantees sustained AI performance.

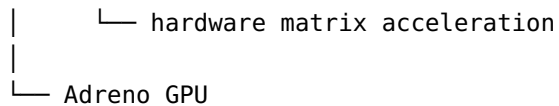
The Snapdragon AI engine under the microscope

This is the most important part for your project.

Snapdragon 8 Elite Gen 5's AI system is not one single block. Qualcomm describes an AI Engine combining:

Qualcomm AI Engine





These NPU features are explicitly listed in Qualcomm's current Snapdragon 8 Elite Gen 5 specification. ⁹

Qualcomm reports a **37% NPU performance improvement** and **16% better NPU performance per watt** versus the preceding generation, and it adds INT2 and FP8 support in addition to other inference precisions. Again, those are Qualcomm's comparisons against its prior generation, not independent benchmarks of the iQOO 15 itself. ⁶

For your architecture, the precision support is important because small numerical representations are one of the mechanisms that make multi-billion-parameter models practical on mobile devices.

Conceptually:

```
FP16
16 bits / weight

INT8
8 bits / weight

INT4
4 bits / weight

INT2
2 bits / weight
```

Lower precision can significantly reduce model weight storage and memory traffic, although actual model quality and runtime performance depend on the quantization scheme and which layers are kept at higher precision.

Do not quote an invented mobile TOPS number

Your earlier caution was correct.

Qualcomm's Snapdragon 8 Elite Gen 5 product page lists the architecture and relative NPU improvements but does **not** state a single official mobile NPU TOPS figure. ⁶

This is particularly important because Qualcomm does publish TOPS figures for some *other* products. For example, AI Hub release material explicitly refers to a Dragonwing IQ9075 configuration as 100 TOPS; that is not the Snapdragon 8 Elite Gen 5 phone SoC. ¹⁰

So putting something such as:

“Snapdragon 8 Elite Gen 5 = 100 TOPS”

in your hackathon deck would be unjustified.

Use:

37% faster Hexagon NPU generation-over-generation, according to Qualcomm

instead. ⁶

The Sensing Hub is interesting but shouldn't be a core dependency

Qualcomm specifies a Sensing Hub with a **dual-core AI processor** and always-sensing capabilities. ¹¹

Architecturally, something like this is attractive:

```
Sensing Hub
  ↓
lightweight context
  ↓
personal/local agent
```

But there is a difference between silicon capability and third-party Android API accessibility. Qualcomm's public hardware description does not establish that your normal Android app gets arbitrary access to every internal Sensing Hub AI capability.

Therefore, for the hackathon:

Do not make direct Sensing Hub execution necessary for your MVP.

Build contextual features using ordinary Android APIs first. Treat specialized Sensing Hub access as an enhancement only if iQOO/Qualcomm gives participants an SDK or documented API.

Qualcomm's “220 tokens/sec” claim needs the same caution

Qualcomm's current mobile-AI page advertises on-device generative-AI speeds of “up to 220 token per second” and mentions large models such as gpt-oss 20B. ¹²

That is impressive evidence about the broader capability of Qualcomm's latest mobile AI stack, but it is **not sufficient evidence that your particular iQOO 15 will produce 220 tokens/s with Qwen3-4B or Qwen3-8B**. The page does not associate that headline number with a precise handset, model, quantization, prompt size, context length and sustained thermal state.

Do not put:

iQOO 15 = 220 tok/s

on your slide.

Put:

Qualcomm demonstrates very high on-device LLM throughput on its current mobile AI platform.

Our application measures actual performance on the iQOO 15.

That distinction is excellent engineering practice.

What local model sizes are actually realistic

This is where the deep research meaningfully improves the earlier recommendation.

Qualcomm AI Hub now provides direct evidence that the Snapdragon 8 Elite Gen 5 target is compatible with:

Qwen3-4B — 4-billion-parameter multilingual LLM; Qualcomm lists Snapdragon 8 Elite Gen 5 QRD and Snapdragon 8 Elite Gen 5 Mobile in its supported targets. ¹³

Qwen3-8B — 8-billion-parameter multilingual LLM; Qualcomm likewise lists Snapdragon 8 Elite Gen 5 QRD and the Gen 5 Mobile chipset among supported platforms. ¹⁴

Qwen3-VL-4B-Instruct — a 4B vision-language model that understands text and images, with the Snapdragon 8 Elite Gen 5 QRD and Mobile chipset explicitly listed as supported. ¹⁵

That means your original **3B-8B local range is not unreasonable**. But “supported” and “ideal hackathon choice” are not the same thing.

Raw weight memory gives us a lower bound

For a mathematically ideal 4-bit representation:

Parameter count	Raw 4-bit weights only
1B	~0.5 GB
3B	~1.5 GB

Parameter count	Raw 4-bit weights only
4B	~2.0 GB
7B	~3.5 GB
8B	~4.0 GB
14B	~7.0 GB

Those values are simply parameter count $\times$ 4 bits; they represent **only the theoretical raw weights**.

A running transformer also needs memory for such things as:

```

model weights
+
quantization metadata
+
runtime
+
tokenizer
+
KV cache
+
activations / scratch buffers
+
NPU/GPU buffers
+
Android OS
+
your application
+
camera / UI buffers
+
database/cache

```

Consequently, saying “8B 4-bit = 4 GB, therefore it fits in 12 GB” is too simplistic.

The 12 GB SKU should define your design baseline

The Indian iQOO 15 comes in both **12 GB and 16 GB RAM configurations**. 16

Unless the organizers explicitly guarantee that every hackathon loaner is the 16 GB model, build as though you will receive:

12 GB RAM
256 GB storage

That gives you the safest compatibility target.

I would classify the model ranges this way:

Local model range	Recommendation for your app	Why
sub-1B	Excellent specialist tier	classification, embeddings, safety/routing
1B–3B	Excellent	very comfortable edge reasoning tier
~4B	Primary recommendation	strong balance of capability and headroom
7B–8B	Supported / experimental	useful quality tier, less operating headroom
14B+	Avoid as core MVP	unnecessary memory and thermal risk
20B+	Demo/research territory	don't base hackathon reliability on it
30B–70B	Cloud	wrong problem for phone execution

The last several rows are architecture recommendations rather than Qualcomm compatibility statements.

Four billion parameters is the sweet spot for this specific product

Your local model does **not** need to write all of Bimax.

It needs to understand things such as:

“Team lead assigned you dark mode. Delegate to AI under my supervision.”

and turn it into:

```
{
  "intent": "delegate_task",
  "task_id": "184",
  "executor": "ai",
  "supervisor": "current_user",
  "execution_mode": "supervised"
}
```

It also needs to reason about:

```
Should this be local or cloud?  
Is this high risk?  
Who owns the task?  
Do I need confirmation?  
Which UI should open?  
Can I summarize this diff?  
Is this notification urgent?  
Should we recommend research?
```

Those are far easier than:

```
Read a 2-million-line repo  
+  
understand architecture  
+  
write 800 lines of code  
+  
debug tests  
+  
reason across 100k tokens
```

So using an 8B model merely because the silicon can support one may actually make your product worse through additional startup latency, memory use and thermal load.

The VLM opportunity is particularly valuable for you

Qwen3-VL-4B-Instruct is especially relevant because Qualcomm identifies it as a phone-compatible vision-language model and explicitly lists Snapdragon 8 Elite Gen 5 support. ¹⁵

That could enable your:

```
Camera  
↓  
"What's wrong here?"  
↓  
local VLM  
↓  
"This looks like a UI overflow in  
the checkout modal."  
↓  
project/task mapping  
↓  
cloud coding agent
```

or:

```
Screenshot
↓
local VLM
↓
extract visible error
↓
classify problem
↓
identify project
↓
suggest:
"Investigate with AI"
```

This would make the iQOO materially part of the engineering workflow rather than a glorified notification terminal.

There is one interesting detail: Qualcomm's current page for **Qwen3-VL-2B-Instruct** does not list **Snapdragon 8 Elite Gen 5**, while its 4B VLM page does. That reinforces why you should choose models based on the actual Qualcomm deployment matrix rather than assuming "smaller automatically means better supported." ¹⁷

The Android runtime stack is much stronger than it first appears

The practical question is no longer:

"Can Snapdragon theoretically run AI?"

It is:

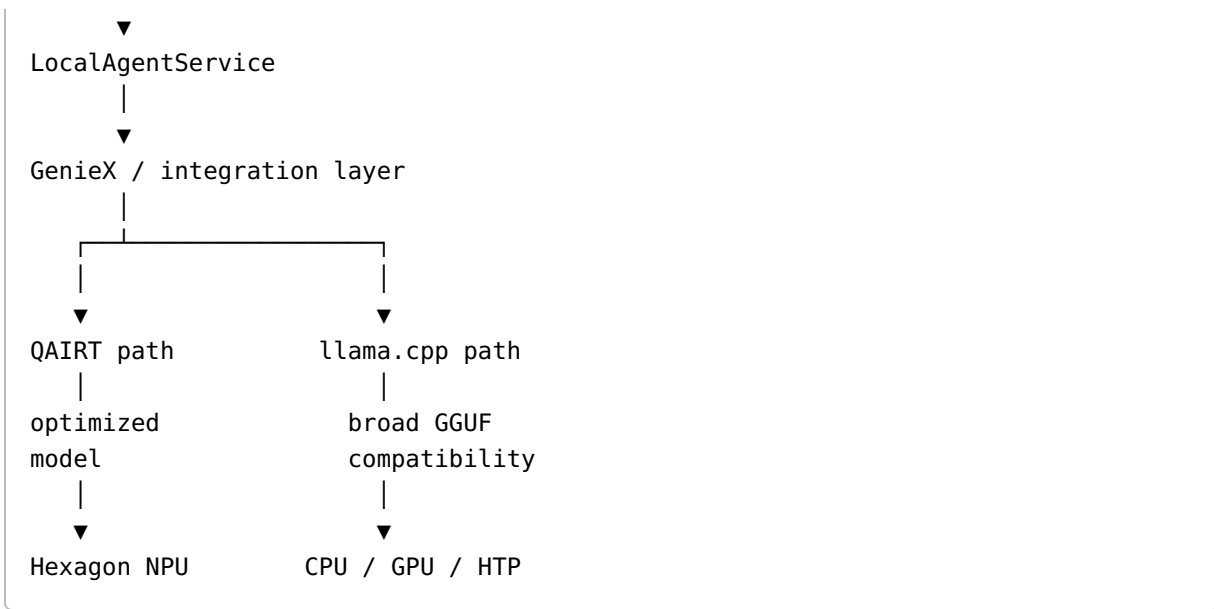
"How do we actually put the model into our Android application?"

Qualcomm's current answer is increasingly **GenieX + QAI RT / AI Hub**.

Qualcomm describes GenieX as an on-device generative-AI inference runtime supporting language and vision-language models across the **Hexagon NPU, Adreno GPU and CPU**. It exposes a QAI RT plug-in for optimized NPU execution and a llama.cpp plug-in for broader GGUF model coverage. Qualcomm also lists Java APIs/SDK support, which is particularly relevant to an Android/Kotlin application. ¹⁸

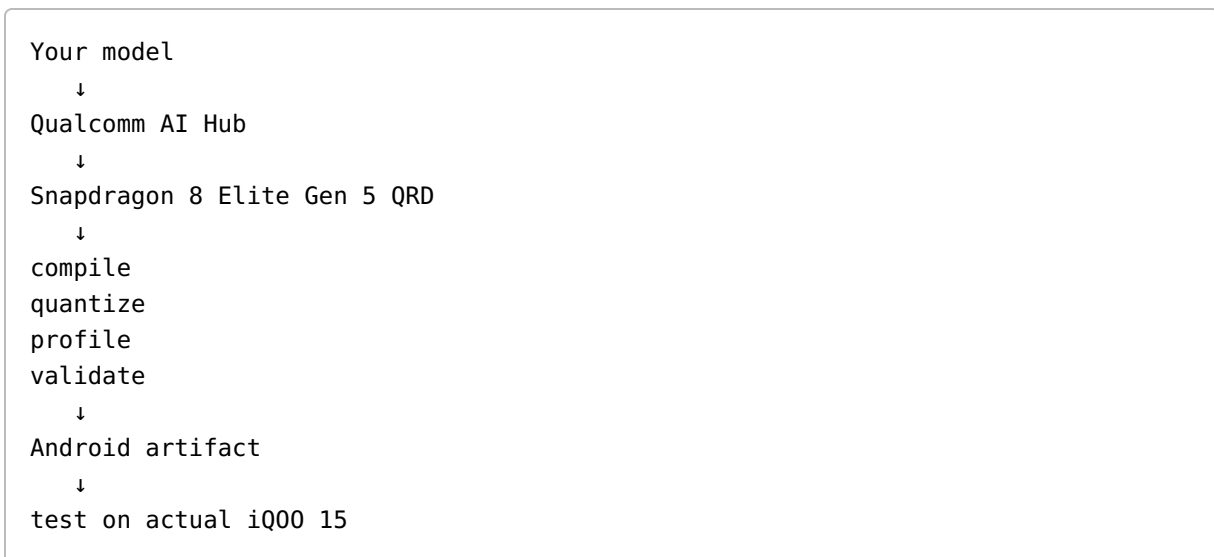
The conceptual stack becomes:

```
YOUR KOTLIN APP
|
```



Qualcomm AI Hub Workbench can take PyTorch or ONNX models, convert/compile them for Qualcomm runtimes, quantize them and profile them on hosted Qualcomm devices. As of September 2026, Qualcomm's release notes show active development through **QAI RT 2.49**, and the Snapdragon 8 Elite Gen 5 QRD has been available as an AI Hub target since September 2025. ¹⁹

That means you can start optimization **before physically getting the hackathon phone**:



This is a major practical advantage. Qualcomm explicitly documents targeting the Snapdragon 8 Elite Gen 5 QRD through AI Hub. ²⁰

But QRD benchmark numbers are not iQOO 15 benchmark numbers

Qualcomm explicitly warns developers that results from similar/proxy devices can differ on an actual commercial handset because of:

- OS
- firmware
- clock speeds
- memory
- thermal packaging
- other device-specific factors. ²¹

That point is extremely relevant to you.

You can optimize against:

Snapdragon 8 Elite Gen 5 QRD

before the event.

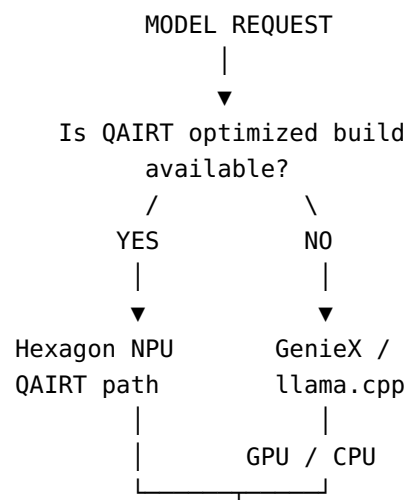
But once you receive:

iQOO 15

you should immediately profile the actual device.

Recommended runtime priority

For your hackathon build, I would use:



↓
cloud fallback

For known Qualcomm-optimized models such as Qwen3-4B, I would prefer **QAIRT/NPU execution**. ²²

For some experimental community model you suddenly want during the hackathon, GGUF/llama.cpp gives you a much easier compatibility route, even if it may not deliver the same NPU optimization. ¹⁸

That dual path is excellent hackathon insurance.

The best architecture for your engineering-team product

After examining the hardware and software stack, I would make a stronger architectural distinction than in our earlier version.

The phone should own the control loop

Your iQOO should locally perform:

PERCEPTION

↓

"What did the user say/show?"

INTENT

↓

"What does the user want?"

LOCAL CONTEXT

↓

"Which project/task/person?"

RISK

↓

"Can this happen automatically?"

ROUTING

↓

"Local action or cloud agent?"

SUPERVISION

↓

"What should the user know?"

APPROVAL

↓

"Does the human authorize this?"

That is meaningful on-device AI.

The cloud then owns expensive execution.

Local task matrix

Capability	Where I would run it	Reason
Voice command understanding	Phone	immediate interaction
Intent → structured action	Phone	low latency, cheap
Task classification	Phone	ideal small-model job
Notification prioritization	Phone	personal/private context
Diff summary for moderate diff	Phone first	cheap and fast
Task/person recommendation	Phone + server data	local reasoning over retrieved context
Risk classification	Phone + deterministic policy	useful but policy remains authoritative
Local project-memory retrieval	Phone	privacy + low latency
Screenshot triage	Phone	strong NPU/VLM use case
Camera → engineering issue	Phone first	showcases phone hardware
Approval explanation	Phone	immediate
Repo-wide coding	Cloud	context/tool workload
Building / tests	Cloud sandbox	needs filesystem/tooling
Deep web research	Cloud	network/search heavy
Full PR review	Cloud	large code context
macOS DMG compilation	macOS cloud runner	platform-specific
Production deployment	Backend control plane	security/auditability

The local side is technically plausible because Qualcomm now explicitly supports language and vision-language execution on Hexagon/Adreno/CPU and lists 4B/8B models for the Gen 5 target. ²

The critical security distinction

The model on the phone can recommend:

```
{  
  "risk": "high",  
  "recommended_action": "require_biometric_approval"  
}
```

But the model should **not itself define security policy**.

Your deterministic application/backend policy still says:

```
production deploy    → biometric approval  
rollback             → biometric approval  
database migration   → biometric approval  
secret modification → biometric approval  
DNS modification     → biometric approval
```

That way hallucination cannot bypass authorization.

So:

```
AI = judgment / recommendation  
  
Policy engine = authority
```

Your exact dark-mode example becomes more interesting

Imagine the team lead sends:

"Sid, add dark mode to Bimax."

Your iQOO receives task metadata.

The local model sees:

```
project = Bimax  
assigned_to = Sid  
priority = normal  
task = add dark mode
```

The phone itself can generate:

Team Lead assigned you “Add Dark Mode.”

This looks suitable for supervised AI implementation.

[I'll build it] [AI assist] [Delegate to AI]

No cloud LLM needs to be called merely to understand that screen.

You select:

Delegate to AI under my supervision

The phone locally converts that to:

```
{
  "action": "delegate",
  "task_id": "184",
  "execution_mode": "supervised_ai",
  "human_owner": "sid"
}
```

Then:

```
iQOO 15
|
| HTTPS
▼
Cloudflare Worker
|
▼
Node Orchestrator
|
├─ GitHub App
├─ NVIDIA NIM
└─ Vercel Sandbox
```

The cloud coding agent changes Bimax.

Later it returns:

```
12 files changed
42 tests passed
1 concern
```

The phone's local LLM can turn the structured results and selected diff chunks into:

AI added system/light/dark themes, persisted the preference and updated six components. One hard-coded terminal background remains.

Then it can ask locally:

Would you like a reviewer agent to investigate that concern?

That is precisely the sort of interaction where a 4B local model is highly valuable.

Cloud escalation should be explicit and visible

A particularly strong hackathon UI would show:

```
LOCAL AI
✓ understood request
✓ found current task
✓ classified risk
✓ prepared action

CLOUD AI
● implementing feature
```

Later:

```
LOCAL AI
✓ summarized result
✓ ranked concern

CLOUD AI
✓ reviewer inspected full repository
```

This visually proves to judges that the iQOO is contributing computationally rather than merely calling an API.

Research can also be locally initiated

Suppose a diff contains a library/API the phone's model does not recognize confidently.

The local model can output:

```
{
  "confidence": 0.58,
  "uncertainty": "framework API behavior",
  "recommended_tool": "web_research"
}
```

Then the UI says:

This change depends on behavior I'm not confident about. I recommend checking the latest official documentation.

[Research]

Only then does your cloud research agent run.

That makes your architecture:

```
Local intelligence
    ↓
detect uncertainty
    ↓
request external evidence
    ↓
Cloud research
    ↓
evidence returned
    ↓
local supervision UI
```

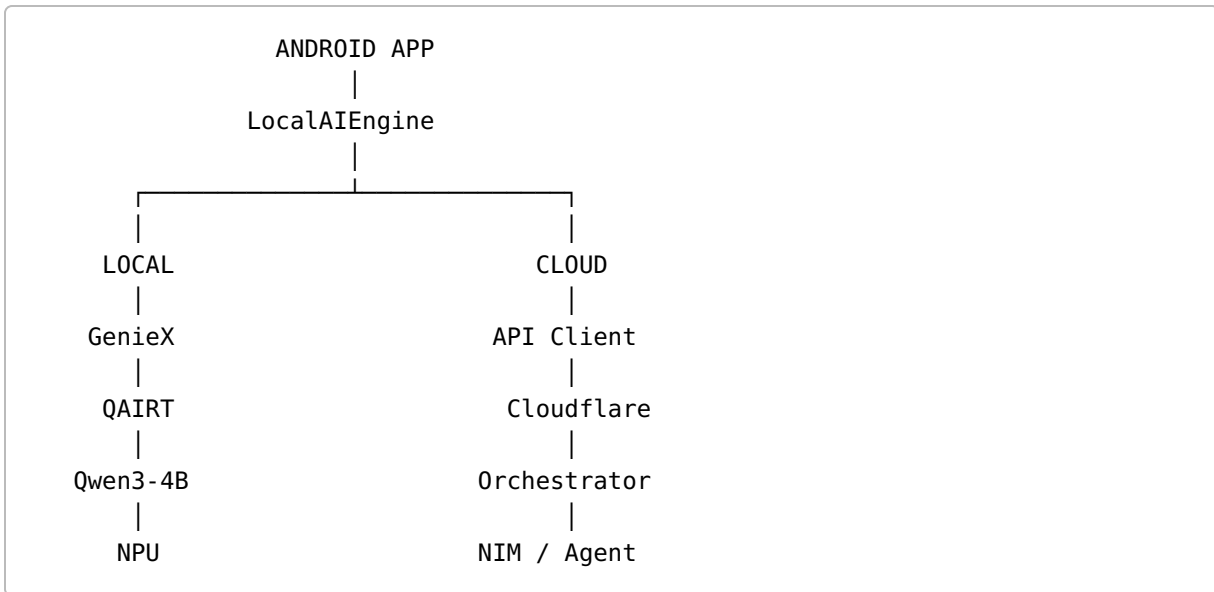
This is far more interesting than simply wiring every user message directly into a huge cloud model.

Hackathon implementation and validation strategy

The smartest route is **not** to arrive at the event and begin figuring out Snapdragon inference from zero.

Qualcomm AI Hub has supported Snapdragon 8 Elite Gen 5 model testing since September 2025, and its current stack supports model optimization, quantization, deployment and profiling. ²³

I would prepare the following architecture before receiving the actual handset:



Primary model choice

My first target would be a **Qwen3-4B family model through Qualcomm's hardware-optimized QAIRT path**, because Qualcomm currently lists Qwen3-4B as hardware optimized and explicitly supports Snapdragon 8 Elite Gen 5. ²⁴

I would constrain it heavily:

Context:
recent task + relevant project metadata

Output:
structured JSON when controlling app

Temperature:
low for actions

Tools:
only local safe tools

Responsibilities:
intent
routing
summaries
delegation
explanations
risk suggestions

Do not give it a 50,000-token chat history simply because an LLM technically allows large context.

Your local experience should feel instantaneous and bounded.

Multimodal option

For camera/screenshot features, test **Qwen3-VL-4B-Instruct** because Qualcomm explicitly lists Snapdragon 8 Elite Gen 5 among its supported targets. ¹⁵

A strong demo could be:

```
You photograph a broken Bimax screen
      ↓
iQOO local VLM
      ↓
"It appears the sidebar text has
insufficient contrast in dark mode."
      ↓
Find Bimax project
      ↓
Create issue/task
      ↓
"Delegate to AI?"
      ↓
cloud coding agent
```

Now the phone's camera, NPU, local multimodal model, team-control system and cloud coding architecture all participate in one coherent interaction.

That is a much stronger demonstration than:

"Here's our chatbot."

Secondary 8B experiment

Qwen3-8B is officially listed as supported on Snapdragon 8 Elite Gen 5, so it is absolutely worth benchmarking. ¹⁴

But make it:

OPTIONAL HIGH-QUALITY MODE

rather than your baseline.

Test:

4B
vs
8B

for your actual workflows:

- command classification
- structured delegation
- diff summarization
- technical explanation
- routing
- local question answering.

If 4B gets essentially the same answers while responding more quickly and leaving more RAM free, use 4B.

What to measure as soon as you receive the loaner

Do not benchmark only “tokens per second.”

Measure:

Metric	Why it matters
Model load time	first-use UX
Time to first token	perceived responsiveness
Decode tokens/sec	generation speed
Peak process memory	stability
10-minute sustained inference	thermal behavior
30-minute mixed workflow	realistic demo load
battery drop	edge-AI energy cost
device thermal status	throttling
NPU placement	prove hardware acceleration
structured-output success rate	agent reliability
cloud fallback rate	architecture efficiency

Qualcomm AI Hub supports profiling and NPU-segment inspection, while its documentation also cautions that real-device results can differ from reference/proxy hardware because of OS, firmware, clocks, memory and thermal packaging. ²⁵

Your eventual presentation can then contain **your own iQOO 15 measurements**:

ON-DEVICE AI

Model: Qwen3-4B [exact quantization]
Hardware: Snapdragon 8 Elite Gen 5
Runtime: QAI RT
Device: iQ00 15

Task routing latency:	XXX ms
TTFT:	XXX ms
Generation:	XX tok/s
Peak memory:	X.X GB
Cloud calls avoided:	XX%

Those figures would be enormously more convincing than repeating a generic TOPS claim.

Design for the 12 GB device

Until organizers explicitly state otherwise:

Target:
12 GB RAM

Treat:
16 GB as bonus headroom

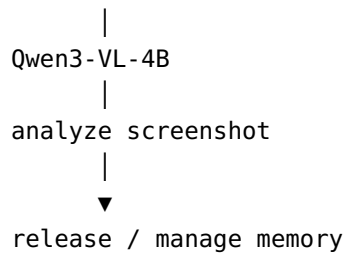
The official India specifications confirm both capacities, so this is the conservative engineering choice. ¹⁶

I also would **not keep multiple multi-billion-parameter models resident simultaneously** on the 12 GB configuration unless testing proves it safe.

Instead:

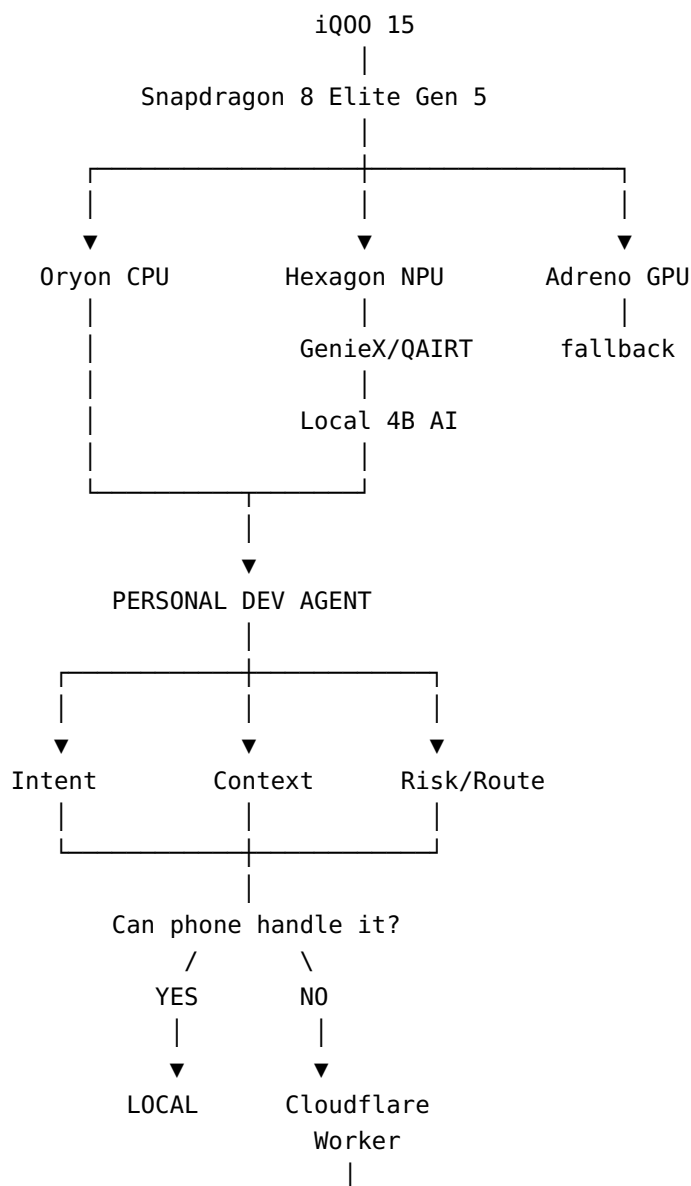
```
Normal operation
Qwen3-4B text model
|
├─ intent
├─ summaries
├─ routing
└─ delegation

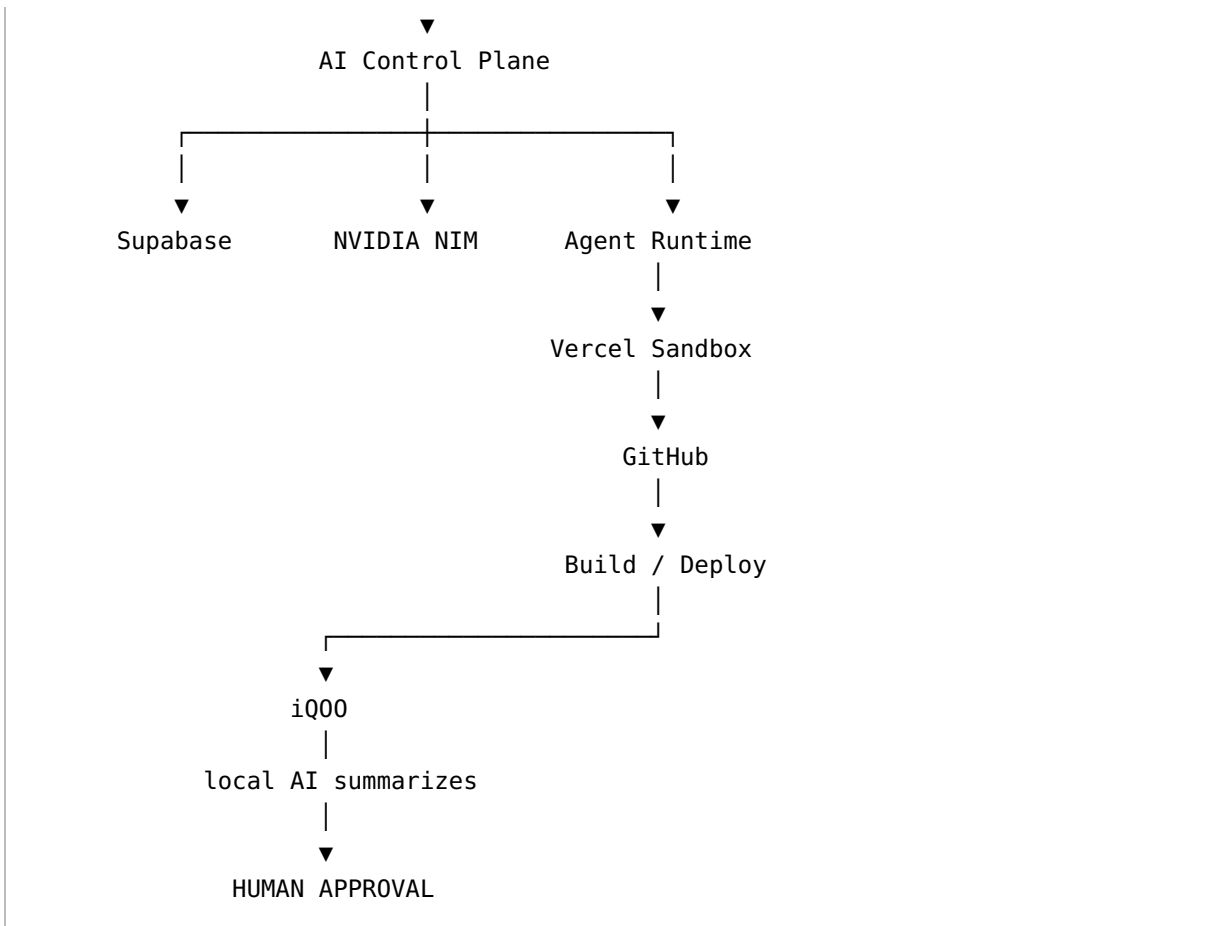
Need visual analysis?
|
▼
transition / load VLM
```



Or use a significantly smaller dedicated vision pipeline for common cases and reserve the VLM for ambiguous screenshots.

The final architecture I would pitch





That gives every layer a real reason to exist.

The **iQOO** performs perception, personal context, reasoning, routing and supervision.

The **NPU** is genuinely used.

The **cloud model** handles tasks that justify cloud compute.

The **sandbox** actually manipulates code.

The **control plane** enforces policy.

The **human** remains the final authority for risky operations.

Final assessment

After the deeper verification, I would revise our previous capability chart to this:

Workload	iQOO 15 assessment
Intent detection	● Excellent
Structured commands / JSON	● Excellent
Notification summarization	● Excellent
Task routing	● Excellent
Risk recommendation	● Excellent
Local RAG	● Excellent
Embeddings	● Excellent
Small classifiers	● Excellent
1B–4B local LLM	● Primary target
4B local VLM	● Supported by Qualcomm stack; test actual device
Screenshot analysis	● Strong use case
Camera → bug triage	● Strong use case
7B–8B quantized LLM	● Technically supported; benchmark before depending on it
Long 8B sessions	● Depends on context/thermal/memory behavior
Full-repository coding	● Cloud
Huge context code reasoning	● Cloud
14B+ as always-loaded agent	● Poor MVP choice
30B–70B frontier coding model	● Cloud
macOS application build	● macOS cloud runner

Qualcomm's current model catalog is particularly important evidence here: both Qwen3-4B and Qwen3-8B list Snapdragon 8 Elite Gen 5 Mobile support, while Qwen3-VL-4B-Instruct provides a directly relevant multimodal path for your screenshot/camera features. ²⁶

The most technically credible hackathon claim is therefore **not**:

“We run the entire engineering organization on a phone.”

It is:

“The iQOO 15 is the edge intelligence and trust layer for an AI engineering team. Snapdragon's NPU handles perception, local reasoning, team context, routing and supervision on-device; expensive repository-scale coding and research are selectively escalated to cloud agents.”

And the architecture can demonstrate that claim live:

SEE
camera / screenshot / task



UNDERSTAND
local NPU model



RETRIEVE
local team/project context



DECIDE
intent + risk + routing



ACT
local action
or
cloud agent



SUPERVISE
local explanation



AUTHORIZE
human + biometric approval

That is a substantially stronger use of the iQOO 15 than treating an extremely capable Snapdragon device as a display for an API call. It aligns the phone's verified local AI capabilities with the phone-first nature of the current iQOO Hackathon while keeping the engineering architecture realistic about what belongs in the cloud. ²⁷

1 16 iQOO 15 Launched

<https://www.iqoo.com/in/products/param/iqoo15>

2 18 GenieX - Qualcomm AI Hub

<https://aihub.qualcomm.com/geniex>

3 19 24 Qualcomm AI Hub

<https://aihub.qualcomm.com/>

4 6 Snapdragon 8 Elite Gen 5 | Snapdragon

https://www.qualcomm.com/smartphones/products/8-series/snapdragon-8-elite-gen-5?utm_source=chatgpt.com

5 9 11 Snapdragon 8 Elite Gen 5 | Snapdragon

<https://www.qualcomm.com/smartphones/products/8-series/snapdragon-8-elite-gen-5>

7 8 iQOO 15 | iQOO Global

<https://www.iqoo.com/in/products/iqoo15>

10 20 21 23 25 Release Notes — Qualcomm® AI Hub documentation

https://workbench.aihub.qualcomm.com/docs/hub/release_notes.html

12 Mobile AI Solutions | On-Device AI Benefits | Qualcomm

https://www.qualcomm.com/smartphones/features/mobile-ai?utm_source=chatgpt.com

13 22 26 Qwen3-4B - Qualcomm AI Hub

https://aihub.qualcomm.com/models/qwen3_4b

14 Qwen3-8B - Qualcomm AI Hub

https://aihub.qualcomm.com/models/qwen3_8b

15 Qwen3-VL-4B-Instruct - Qualcomm AI Hub

https://aihub.qualcomm.com/models/qwen3_vl_4b_instruct

17 Qwen3-VL-2B-Instruct - Qualcomm AI Hub

https://aihub.qualcomm.com/models/qwen3_vl_2b_instruct

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https://iqoo.reskill.com/?utm_source=chatgpt.com